SIH26173 iTantra — Referenced Material and Research Links

Offline, device-to-device voice messaging for Indian languages (STT → compact sealed text → TTS over Bluetooth / Wi-Fi Direct).

Organisation: Indian Space Research Organisation (ISRO), Department of Space.
Category: Software.

Compiled: 30 September 2026. This document lists the primary source documents, research papers, models, datasets, standards and prior art behind the project, with links. Open every link before citing it on a slide.

1. Local source documents (in this workspace)

Document	Location	What it is
iTantra research, gap analysis and build plan (PDF)	/home/nikhil/dev/SIH/SIH26173 iTantra research, gap analysis and build plan-1.pdf	The main research doc: findings, gap analysis, measured bandwidth, model/licence notes, what not to claim. Referenced throughout.
Agent development plan	/home/nikhil/dev/SIH/ITANTRA_AGENT_PLAN.md	The ordered build plan, contract, phases and gates used for the implementation.
Repository README / TASKS / DECISIONS / OPEN_QUESTIONS	/home/nikhil/dev/SIH/itantra/	Implementation status, evidence log, decision log and blockers.
Protocol, packs, metrics, licences, demo	/home/nikhil/dev/SIH/itantra/docs	PROTOCOL.md, PACKS.md, METRICS.md, LICENSES.md, DEMO.md, PS_OFFICIAL.md.
Measured results	/home/nikhil/dev/SIH/itantra/tools/bench/results	size.csv, link.csv and their provenance sidecars. Every number quoted must come from here.

2. Problem statement (summary)

Verified from listings: PS number SIH26173; Title 'iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links'; Organisation ISRO; Category Software. Detailed requirements circulating in team repos are labelled 'reported' and must be checked against the portal copy.

Reported scoring: Efficiency 20%, Accuracy 40%, Latency 20% (the fourth 20% is not stated in the sources examined). Reported languages: ten (Hindi, Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Bengali, Odia, Punjabi) plus English.

Official portal: https://sih.gov.in/

Note: the official portal text could not be opened during research; requirements marked 'reported' come from third-party team repositories that quote it.

3. Research papers (verified links, open before citing)

Paper	Link	Why it matters here
Polyak et al., Speech Resynthesis from Discrete Disentangled Self-Supervised Representations (Interspeech 2021)	arxiv.org/abs/2104.00355	A 365 bps semantic codec (content + pitch + speaker). Use as the 'semantic codec' comparison for text bitrate; not Indic, needs HuBERT and a GAN vocoder.
Pratap et al., Scaling Speech Technology to 1,000+ Languages (MMS, Meta)	arxiv.org/abs/2305.13516	Source of the MMS-TTS voices (CC-BY-NC 4.0) used for languages Piper lacks. Non-commercial banner required.
Javed et al., IndicVoices: an Inclusive Multilingual Speech Dataset for Indian Languages (ACL Findings 2024)	arxiv.org/abs/2403.01926	7,348 h, 22 languages, spontaneous speech. Justifies testing on spontaneous/noisy speech, not only read sentences.
Sankar et al., IndicVoices-R (NeurIPS 2024)	arxiv.org/abs/2409.05356	1,704 h TTS-grade speech, 10,496 speakers, 22 languages. Only if fine-tuning a missing voice.

Gala et al., IndicTrans2 (TMLR 2023, AI4Bharat)	arxiv.org/abs/2305.16307	Translation for all 22 scheduled languages; 1.1B base, ~211M distilled. Optional stretch; phone speed unproven.
Leschanowsky et al., Benchmarking Neural Speech Codec Intelligibility with SITool (Interspeech 2025)	arxiv.org/abs/2506.01731	Subjective intelligibility benchmark for AMR/EVS/Codec2/LPCNet /Lyra V2/Mimi; method for a small listening test of TTS output.
Gulati et al., Conformer (Interspeech 2020)	arxiv.org/abs/2005.08100	Architecture behind IndicConformer.
Kim et al., VITS (ICML 2021)	arxiv.org/abs/2106.06103	Architecture behind Piper voices (end-to-end TTS).
Radford et al., Whisper (2022)	arxiv.org/abs/2212.04356	Baseline multilingual ASR; one team reports ~100% WER on a single Tamil sentence (weak evidence).
Conneau et al., FLEURS: Few-shot Learning Evaluation of Universal Representations of Speech	arxiv.org/abs/2205.12446	Public multilingual test set (reported CC-BY-4.0); verify licence before use.
Hsu et al., HuBERT	arxiv.org/abs/2106.07447	Self-supervised units used by semantic codecs.
Baevski et al., wav2vec 2.0	arxiv.org/abs/2006.11477	Pre-training base for MMS and Indic models.

4. Papers cited inside the research PDF (verify the link before citing)

These appear in the research PDF's section 3; the PDF gives identifiers, not open URLs. Confirm each resolves before quoting.

Work	Identifier / link	Relevance
STCTS: speech as ~80 bps text plus sparse prosody	arXiv:2512.00451	Same idea as ours; not Indic, not on phones.
Collette, Greenwood, Nicoll: semantic	arXiv:2509.15462	Semantic compression evidence; not a phone

compression matches codecs at 2–4× lower bitrate		system.
LMCodec2: sub-1 kbps voice for Beidou-3 short messages	ScienceDirect 2024 (search 'LMCodec2 ScienceDirect 2024')	Real narrow-link use case for very low bitrate voice.
BhasaAnuvaad: read vs spontaneous speech	arXiv:2411.04699	Justifies testing on spontaneous/noisy speech.
Rasmalai / Indic Parler-TTS	arXiv:2505.18609	Server-class, 32×H100; do not pitch as on-device.
On-device streaming ASR with post-training quantization	arXiv:2604.14493	2.47 GB → 0.67 GB within 1% absolute WER (English); supports quantizing Indic models.
SN-WER: script-normalized error rate for multi-script Indic ASR	arXiv:2606.02548	Report CER and WER and note multi-script effects.

5. Models, runtimes and tools

Item	Link	Notes / licence to record in the pack manifest
AI4Bharat IndicConformer (per-language hybrid CTC-RNNT)	huggingface.co/ai4bharat/indicconformer_stt_hi_hybrid_ctc_rnnt_large	MIT on the model card; ~120M encoder, 17 Conformer blocks. Third-party int8 CTC ONNX export ~140 MB. Check licence per file.
AI4Bharat IndicConformer 600M multilingual	huggingface.co/ai4bharat/indic-conformer-600m-multilingual	Too heavy for low-end phones; edge-laptop reference only.
Piper (VITS) voices	github.com/rhasspy/piper	Repo MIT; each voice's dataset licence differs (e.g. Marathi CC-BY-SA 4.0). int8 ~21 MB for some voices.
sherpa-onnx (VAD / ASR / TTS runtime,	github.com/k2-fsa/sherpa-onnx	Vendor one release and record the tag. Used for

Kotlin JNI)		Silero VAD, CTC ASR and VITS TTS.
Silero VAD	github.com/snakers4/silero-vad	Voice activity detection; verify frame size from the vendored release.
Meta MMS-TTS	arxiv.org/abs/2305.13516	CC-BY-NC 4.0 (non-commercial). Convert to ONNX per voice.
AI4Bharat models home	ai4bharat.iitm.ac.in	IndicConformer, Indic-TTS, IndicTrans2, IndicVoices.
Google Tink (crypto)	github.com/tink-crypto/tink-java	Ed25519, X25519, HKDF, ChaCha20-Poly1305; or libsodium.
Protocol Buffers	protobuf.dev	Baseline packet format used by other teams; ours is a fixed binary format.

6. Datasets and test corpora

Dataset	Link	Licence / use
FLEURS	arxiv.org/abs/2205.12446	Reported CC-BY-4.0; verify. Read-speech ASR test set.
IndicVoices	arxiv.org/abs/2403.01926	Spontaneous Indic speech; check access terms and test-set availability.
IndicVoices-R	arxiv.org/abs/2409.05356	Indic TTS dataset.
Mozilla Common Voice	commonvoice.mozilla.org	CC0; noisy/read multilingual speech.
In-repo seed corpus	/home/nikhil/dev/SIH/itantra/tools/bench/corpus/seed.jsonl	Hand-authored sample data for the size pipeline; labelled 'sample data', not a benchmark.

7. Communication technologies and standards

Topic	Link	Use here
Bundle Protocol v7 (RFC	rfc-editor.org/rfc/	Store-carry-forward ideas

9171)	rfc9171.html	for the outbox; do not claim BPv7 compliance.
Opus (RFC 6716)	rfc-editor.org/rfc/rfc6716.html	Audio baseline (~6 kbps) for the size comparison.
Codec2	github.com/drowe67/codec2	700–3200 bps voice baseline; 700C is not actively maintained.
Lyra V2	github.com/google/lyra	3.2–9.2 kbps neural codec on Android/TFLite; second audio baseline.
ChaCha20-Poly1305 (RFC 8439)	rfc-editor.org/rfc/rfc8439.html	AEAD for message framing.
Ed25519 (RFC 8032)	rfc-editor.org/rfc/rfc8032.html	Device identity signing.
X25519 (RFC 7748)	rfc-editor.org/rfc/rfc7748.html	Ephemeral key agreement.
HKDF (RFC 5869)	rfc-editor.org/rfc/rfc5869.html	Session key derivation.
Bluetooth RFCOMM (Android)	developer.android.com/.../BluetoothSocket	First demo transport; stream framing over the socket.
Wi-Fi Direct / Wi-Fi P2P (Android)	developer.android.com/.../wifip2p	Transport named by the PS; group owner runs the TCP server.
Nearby Connections	developers.google.com/nearby/connections/overview	Optional fallback (~100 m, picks BT/Wi-Fi).
Foreground service types (Android 14+)	developer.android.com/.../fgs/service-types	microphone + connectedDevice types and permissions.
Meshtastic (LoRa)	meshtastic.org	Stretch: honest narrow-link (LoRa) demo via its AIDL service.

8. Existing systems / prior art

System	Link	What it means for us
Google Translate offline (Live Translate)	support.google.com/translate	On-device translation, Pixel 9–11 only; Hindi is the only Indic language. No device-to-device

		transport.
Bhashini (Government of India)	bhashini.gov.in	ASR/TTS/translation APIs, 22 languages; cloud-first, excluded by 'offline'.
Briar	briarproject.org	Delivery-layer prior art: syncs over Bluetooth/Wi-Fi; no speech.
Meshtastic + ATAK plugin	meshtastic.org	Closest transport prior art: STT (Vosk) over LoRa + TTS on receiver; English only.
AI4Bharat models	ai4bharat.iitm.ac.in	Components (ASR/TTS/MT) with no communication layer.
Codec 2 / FreeDV	github.com/drowe67/codec2	The narrow-link voice baseline to beat.
STCTS (preprint)	arXiv:2512.00451	Same text+prosody idea; not Indic, not on phones.

9. Our measured results (from this implementation)

Every number below is produced by tools/bench or the test suites; the files are in tools/bench/results and the evidence is in TASKS.md.

Result	Value	Source file
Script-aware coding vs UTF-8	937 B vs 2117 B over 28 sentences (avg 33 B/sentence)	tools/bench/results/size.csv
Preset alerts	3 bytes each (codebook reference)	tools/bench/results/size.csv
Reliable vs fire-and-forget at 30% loss (1 kbps)	100% vs 69%	tools/bench/results/link.csv
Duplicate receipts absorbed; duplicate playbacks	13,068 absorbed; 0 playbacks (dedupe window)	tools/bench/results/link.csv + ProtocolSessionTest
Test suites	59 Kotlin JVM, 42 TypeScript, 7 web, 5 Playwright	TASKS.md
Release APK	40.5 MB, 0 INTERNET, signed CN=iTantra	apps/mobile/android/.../app-release.apk

10. Claims to avoid (from the research PDF, section 16)

- Do not claim a WER/CER without a results row for that language, device and condition.
- Do not say 'supports all ten languages' unless all ten are verified; state which are experimental.
- Do not say '98% smaller than audio' without naming the baseline; against Codec2 700C the gain is much smaller.
- Do not claim 'no internet' until the release manifest is verified to have no INTERNET permission (we strip it; verify).
- Do not claim zero latency or real-time voice; report measured, labelled latency.
- Do not claim range in metres, mesh or multi-hop unless tested.
- Do not claim perfect recognition or that errors cannot happen.
- Do not claim translation (the core has none) or security beyond the stated handshake.
- Do not claim real-disaster or field validation, or that synthetic-noise results represent real noise.

Reminder: this prototype is build-complete but not field-verified. On-device speech models and the two-phone link must be tested on hardware before any claim.